



Gas detection system

Instructions for Use



WARNING

Strictly follow the Instructions for Use. The user must fully understand and strictly observe the instructions. Use the product only for the purposes specified in the Intended Use section of this document.

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1. Important Information

1.1 For your safety

- Before using this product, carefully read the Instructions for Use.
- Do not dispose of the Instructions for Use. Ensure that they are retained and appropriately used by the product user.
- Only fully trained and competent users are permitted to use this product.
- Comply with all local and national rules and regulations associated with this product.
- Only trained and competent personnel are permitted to inspect, repair and service the product as detailed in these Instructions for Use. Further maintenance work that is not detailed in these Instructions for Use must only be carried out by IGAS or personnel qualified by IGAS.
- IGAS recommends a IGAS service contract for all maintenance activities.
- Properly trained service personnel must inspect and service this product as detailed in the maintenance section of this document.
- Use only genuine IGAS spare parts and accessories, or the proper functioning of the product may be impaired.
- Do not use a faulty or incomplete product, and do not modify the product. Notify IGAS in the event of any component

1.2 Definitions of alert icons

Alert icons are used in this document to provide and highlight text that requires a greater awareness by the user. A definition of the meaning of each icon is as follows:



CAUTION

Indicates a potentially hazardous situation which, if not avoided, could result in physical injury or damage to the product or environment. It may also be used to alert against unsafe practices.



NOTICE

Indicates additional information on how to use the product.

1.3 General Warnings

1. Do not operate the system or its components outside of their rated operating specification.
2. All equipment containing a User Interface must be suitably protected from direct sunlight and rain.
3. Power Supply Fluctuations are not to exceed AC 180 – 250V Supply or $\pm 10\%$ of nominal.
4. All versions of Enclosure apparatus are electrical Class 1, and must be connected to Protective Earth (Ground).
5. The GDS installation must include a means of isolating or disconnecting the input voltage supply. The isolation or disconnection device must be conveniently located and clearly labelled. AC mains voltage supply disconnection device must disconnect both the line and neutral poles, but maintain earth (ground) continuity.
6. Input voltage supply must include over-current protection.
7. All cabling must be appropriately rated and approved in accordance with local, national and company regulations, and suitable for the installation.

Additionally, cabling must satisfy requirements defined in the manuals of connected field devices, in particular if the field device is certified for use in a hazardous location.

8. Cable entry glands, blanking plugs, reducers, adaptors and breather devices must be suitably approved and must not reduce the IP rating or protection levels. Items should not be used if there is a high risk of mechanical damage to the equipment or enclosure.
9. Access doors and entry points must be kept closed when the system is energised in normal operation.
10. GDL contain hazardous live terminals. Appropriate precautions should be taken during operation, installation, and maintenance and servicing. Specifically, operators must have appropriate training and experience to be aware of the hazards to which they may be exposed, and of measures to minimise risk to themselves or other people.
11. The protection provided by GDS may be impaired or lost if the equipment is installed or used in an incorrect, unspecified or unauthorised way.
12. Exposure to some chemicals may degrade the sealing properties of materials used in the alarm relay.

1.4 General Cautions

1. IGAS CPU, Input and Output Modules have no user serviceable parts. In the unlikely event of a failure, the item must be replaced using only manufacturer supplied parts.
2. Do not use sharp objects to operate the Touchscreen as this could irreparably damage the User Interface and adversely affect its IP rating.
3. Use only soft, damp cloths or screen wipes to clean the GDS. Do not use solvents or abrasives as they will cause irreparable damage.
4. Do not place any objects on top of the enclosures as this may cause overheating and may cause the enclosure to fall from the wall.

1.5 How to Use this Manual

This Manual should be read by everyone who operates or monitors the IGAS gas detection system.

Only personnel who have been fully trained by IGAS are authorised to Install, Set-up, Service, and Test, Repair, or Recondition gas detection systems.

1.6 Conventions Used

Boot up refers to the action of starting the software from cold. *Settings>Service>Relay* highlights a sequence of commands.

Reboot refers to shutting down and restarting the software without interrupting the power supply.

Restart refers to cycling the power off and then on again. [Start] highlights unique button touches.

Start up refers to the action of switching on the system.

Touch refers to all direct interactions with the touchscreen by finger

GDL refers to the Gas Detection System.

2 Safety Cautions and Hazards

2.1 Safety Cautions

Incorrect set-up, maintenance, operation or modification of the GDS or its installation may constitute a serious hazard to the health and safety of personnel and their environment. It is therefore imperative that the contents of this chapter are thoroughly understood by everyone who has access to the gas detection system or its associated equipment.

When properly installed, this gas detection system enclosure is rated as IP65.

In all cases, several hazards may be present when operating or servicing the equipment and extreme caution must be exercised at all times. The hazards that may be encountered include:

- Class 1 electrical hazards (AC 220 V, DC 18–32 V)
- Mechanical hazards (Heavy components, swinging access doors, locking handles etc.)
- Environmental hazards (toxic atmospheres)
- Fire and Ignition hazards

2.2 Safety Hazards

LETHAL VOLTAGE PRESENT

All power supplies must be hard wired and must include a circuit breaker and, close by and unobstructed, a means of manually isolating and locking-out the power supply without breaking the protective earth (ground) connection.

Lethal current may be present in this equipment when electrical power is applied and after it is removed. There is a risk of death or injury from electrical shock when access doors are open.

Isolate power before opening electrical access panels. Ensure residual current is fully discharged before touching live terminals.

Lethal current may be generated both internally and externally to the system. All installations, including enclosures and external units, must be grounded to protective earth, and must be capable of staying earthed (grounded) when the power supply is interrupted.

The GDS and/or its sensors may become contaminated by the ambient environment in which it or they are used. It is the Customer's sole responsibility to ensure that all appropriate safety precautions are taken before handling any components or transferring them to any other party.

RISK OF INJURY AND DAMAGE

Each GDS enclosure is heavy and weighs considerably more when packed. Ensure that a Manual Handling Risk Assessment is carried out before moving or installing the system, and ensure that enclosures are fitted securely to a suitable vertical surface. Do not place objects on the enclosures.

GDS uses high energy AC and DC currents that may cause arcing and sparks if shorted out. Always wear eye protection when the enclosure is open.

Loud alarms and sirens can be used in system. Always wear hearing protection when working in the vicinity of loud or high-pitched noises.

Always follow a Safe System of Work when carrying out any work involving Safety Systems.

2.3 Component Testing and Replacement

Before carrying out any electrical testing or component replacement:

- Read this Manual to become familiar with the location of high voltage components.
- Isolate the system at the main circuit breaker, lock it in the 'Off' position, and attach a notice indicating that maintenance work is in progress.
- Never assume the polarity of cabling or replacement components. Refer to electrical schematics or contact IGAS for confirmation.
- Use only IGAS approved replacement parts.

Only IGAS trained and certified maintenance technicians are authorised to carry out component testing and replacement. Unauthorised work may result in a potentially dangerous situation and will invalidate the manufacturer's warranty.

2.4 Antistatic Precautions

As with all modern electronic circuits, the Printed Circuit Boards (PCBs) in systems utilise some static-sensitive components that can be severely damaged if subjected to static discharge. Static can be generated on the human body by friction or movement and is discharged through the first contacted route to earth. It can also jump gaps between items of differing electrical potential.

2.5 Good Practice

After switching off the system, it is good practice to wait at least 10 seconds before switching it on again. This allows the circuits and RAM to discharge adequately before being powered-up again. Failing to do so may cause data corruption.

2.6 Lithium Battery Hazard

Lithium batteries are fitted to PLC as backup power source. Replace the factory installed battery with type CR2032 only. Use of other batteries may present a risk of fire or explosion.



CAUTION

Lithium batteries may cause severe injury or death if swallowed, and may catch fire or explode if mishandled, recharged, burned or disposed of incorrectly.

Always handle lithium batteries with care, keep them out of the reach of children, and dispose of them carefully in accordance with local regulations.

Batteries shall only be fitted, removed or replaced in non-hazardous (safe) areas.

2.7 Product Compliance

This product complies with the following standards and directives.

Other safety directives may apply to the complete system installation if an OEM's product is integrated into other equipment or machinery.

● **EMC** «TP TC 020/2011»

● **Low Voltage Directive** «TP TC 004/2011»



NOTICE

The Equipment referred to in this manual contains components and assemblies that are each certified for use in a variety of differing environments, and it is the site owner's responsibility to confirm the suitability of the equipment prior to its installation and use.

2.8 Conditions of Use

Before carrying out any electrical testing or component replacement:

This equipment shall only be operated:

- By properly trained personnel.
- Under IGAS approved conditions.
- With due authorisation.
- Using approved maintenance and servicing procedures.

2.9 Training of Personnel

IGAS can provide training for operators and maintenance personnel. Personnel who have been trained in operation and maintenance shall be limited to carrying out only those procedures and tasks taught during the training course. IGAS certified maintenance technicians must carry out all other tasks.

IGAS can also provide additional or advanced training. Retraining is recommended periodically and whenever the equipment / installation is changed or upgraded.

2.10 Due Authorisation

Before any production, maintenance, or servicing procedure is carried out; written authorisation must be obtained from IGAS personnel to confirm that the proposed task satisfies the necessary safety conditions.

2.11 Marking



3 General Description

3.1 System General Description

The Gas Detection System (GDS) is used to continuously monitor remote sensor transmitters in order to warn of potentially dangerous or harmful conditions. System based on a modular PLC intended to be installed in a rack-based system. Other differences are explained in this document where necessary.

GDS can handle a wide range of gas detectors via current loop or RS485 through analogue inputs

It can control various outputs such as audible and visible signals, solenoid valves, venting systems etc. All events transfer via Modbus or IEC60870-5 -104 to control room or SCADA.

Every cabinet has a service USB connector for configuring and diagnostic.

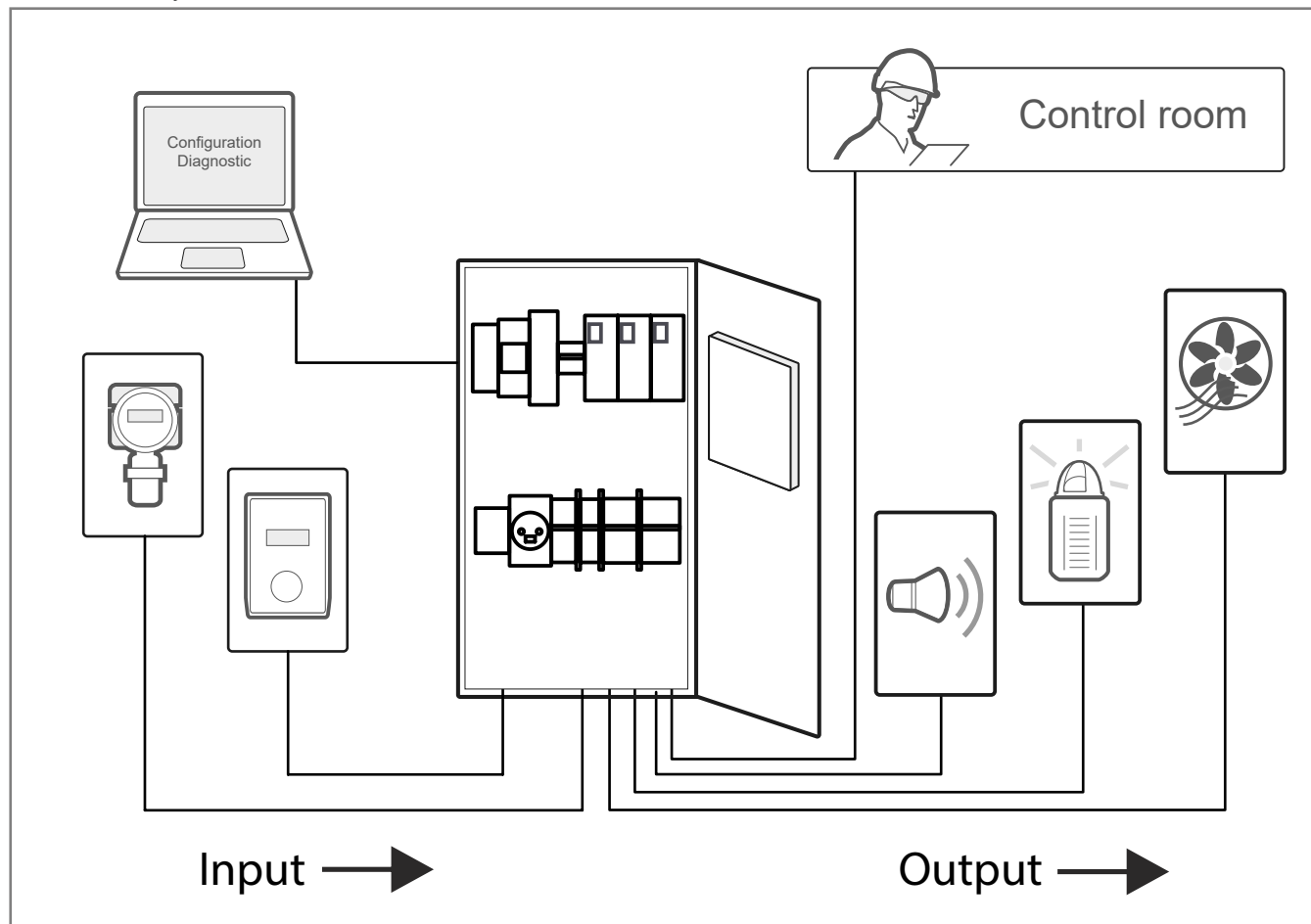


Figure 1. Typical Installation Options



Figure 2. IGAS gas detection system - wall-wounded.

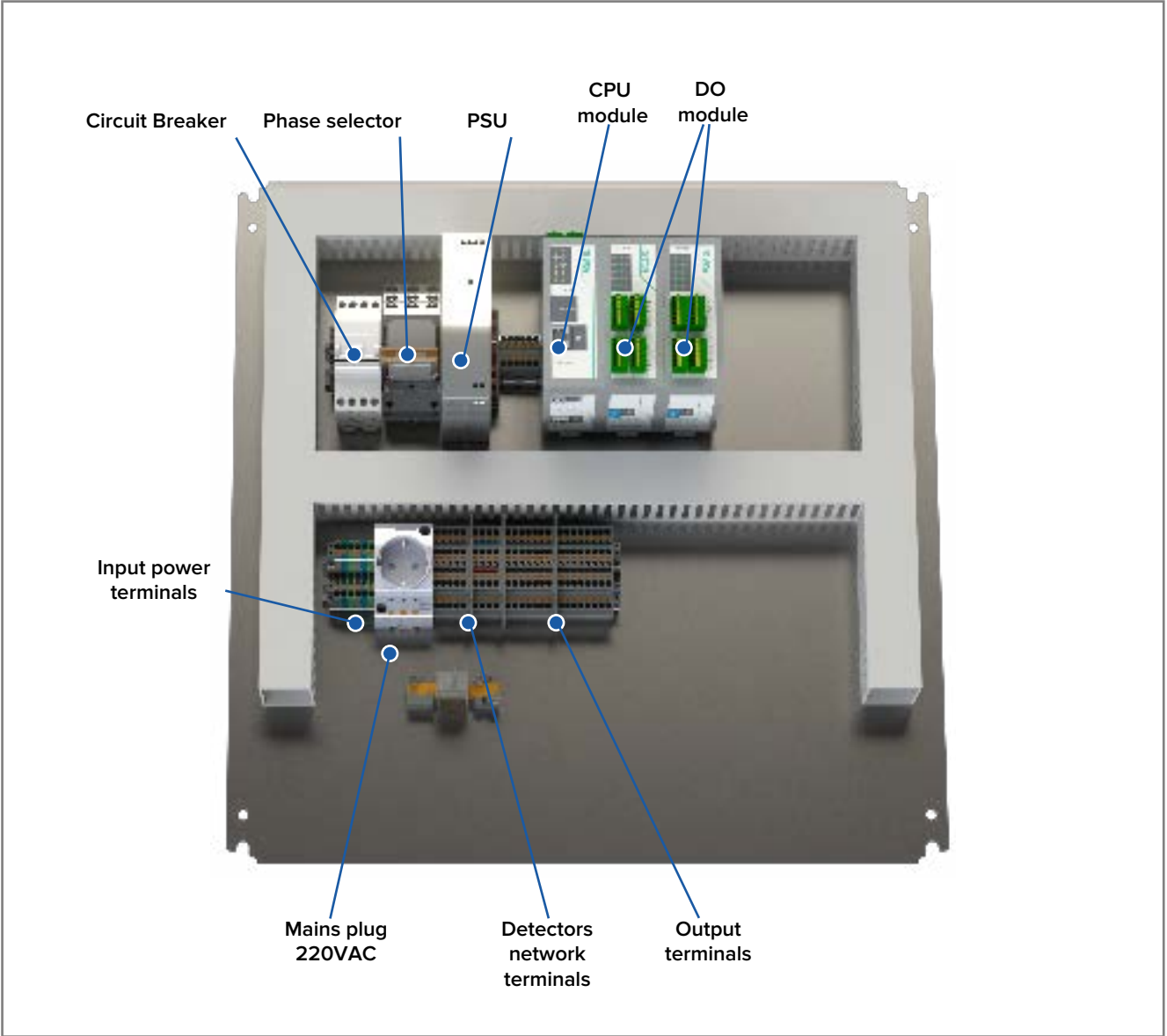


Figure 3. Exploded View

The controller monitors the current signals from the remote sensor transmitters on input channels and provides warnings when alarm or fault conditions occur. Each of the input channels can provide alarms for up to two gas levels and one fault condition. The gas-level alarms are configured to provide an indication of potentially dangerous or harmful gas levels at the transmitter. The fault operate when the transmitter sends the relevant fault signal.

The combination of internal control modules (input modules, relay modules and output modules) and the controller configuration settings can be altered to meet the gas detection and warning requirements.

Additionally the warning devices used with the controller can be selected and configured as required by the user. The GDS is configured using a Microsoft Windows® based software program. The GDS also has a built-in configuration system. The cabinet is constructed from high-impact steel and have fully-sealed, easy opening access. It is supplied with a wall mounting or can be directly mounted to any solid vertical surface. Cable entry is via rubber entry on the lower side. GDS is rated IP65, which means that it is dustproof and can be subjected to low-pressure water without significant ingress. This makes it particularly suited to offices, control rooms and unheated boiler rooms.

Features of the Controller unit:

- Colour LCD Touch Screen with multi-language GUI and menus
- Password protection
- Flexible Mains Power Input: 50 – 60 Hz, 100 ... 240 VAC
- Up to 64 modules on one line
- Remote field bus modules network
- User configurable relay controlled Output
- Alarm update on Acknowledge
- Automatic Self-Diagnostic with error codes
- Event recording
- SD Card
- 2x Ethernet (Modbus TCP, IEC104, PRP)
- Redundant modules

The GDS system is designed to operate on a single phase, 50 to 60 Hz, 100 ... 240 VAC supply with a maximum power consumption of less than 100 W (depends upon configuration). System can contain an optional backup battery to guard against short-term power disruption.

Input Voltage	100...240 VAC
AC Frequency Range	50 – 60 Hz±6%
DC Voltage Output	24 V
Rated Power Output	up to 200W

Table 1. Power Supply Electrical Ratings

Power Usage	Max Power
Touchscreen	5W
CPU module	0.6W
Relay Output Module	1.5W
Analog Input Module	0.3W

Table 2. Power Consumption

3.2 Parameters



NOTICE

Based on one CPU and two Relay output modules

Parameter	Value
External Dimension	
Depth	250* mm
Length	600* mm
Width	600* mm
System weight	25kg*
Ambient Operating Temperature	
standard cabinet	+1 °C to +55 °C
with heater option	-40 °C to +50 °C
Operating Humidity (non-condensing)	5 % to 95 %RH
Storage	
Without batteries	-25 °C to +60 °C 5 % to 95 %RH non-condensing
With batteries	-20 °C to +25 °C -20 °C to +45 °C -20 °C to +60 °C
	1 Year 3 Month 1 Month
IP Rating	IP65

* depends upon configuration

Table 3. System Dimensions

3.3 Construction

The system cabinet are constructed from powder coated steel with a quick release front access door panel.

The GDS door panel holds a touch sensitive colour LCD and 4 status LEDs.

Inside the cabinet is a din rail installed Switched-Mode Power Supply providing a nominal DC 24 V output, an optional Lithium-ion backup battery, a Main CPU Module, two Relay Output Modules, protection circuit breakers, automatic phase selector and input/output terminals.

Enclosure contain a common Earth (ground) rail that must be connected to Protective Earth (Ground) .

3.4 LCP2116 - Processor Module

General information

The module is equipped with:

- 32 bit ARM CPU with integrated I/O
- 2 GB onboard flash drive
- 96 kB SRAM
- Interface for Lorentz modules
- 1 USB interface
- 6 RS485 interfaces
- 2 Ethernet interface 10/100BASE-T
- Push in fast terminals



Figure 4. Processor module

3.5 LCP2116 - Technical data

General information	Discription
Interfaces	6x RS485, 2x Ethernet, 1x USB, 1x X2X network
System module	CPU
Diagnostics	
CPU function	Yes, using status LED
RS485 data transfer	Yes, using status LED
Ethernet	Yes, using status LED
Power supply	Yes, using status LED
X2X network	Yes, using status LED
Controller redundancy possible	Yes
Visual Components support	Yes
Power consumption without interface	1W
CPU and X2X Link power supply	
Input voltage	24 VDC -15% / +20%
Input current	Max. 1 A
Fuse	Integrated, cannot be replaced
Total nominal current	16 A at 30 VDC / 16 A at 250 VAC
Reverse polarity protection	Yes
Controller	
Processor	
Type	ARM
Clock frequency	84 MHz
Memories	
SRAM	100 kB
RAM buffer	4kB ECC
Integrated I/O processor	Processes I/O data points in the background
Remanent variables	512 kB FRAM
Shortest task class cycle time	2ms
Application memory	
Type	2Gb microSD
Writable data amount	40 TB
Guaranteed erase/write cycles	20 000
	IP20

Table 4. Technical Data

Interfaces	
Interface 0	
Signal	RS485
Max. distance	900m
Transfer rate	Max. 1.0 Mbit/s
Connection	2 wire terminal
Interface 1 ... 6	
Signal	RS485
Max. distance	1200m
Transfer rate	Max. 115 kbit/s
Connection	2 wire terminal
Interface 7 ... 8	
Signal	Ethernet
Max. distance	Max. 100 m between 2 stations (segment length)
Transfer rate	10/100 Mbit/s
Connection	1x RJ45 shielded
Transfer	
Physical layer	10BASE-T/100BASE-TX
Half-duplex	Yes
Full-duplex	Yes
Interface 9	
Signal	USB 2.0
Connection	Type B
Electrical properties	
Electrical isolation	Ethernet (IF2), POWERLINK (IF3) and X2X (IF6) isolated from each other Channel isolated from bus
Operating conditions	
Mounting orientation	
Horizontal	Yes
Vertical	Yes
Degree of protection	IP 20
Ambient conditions	
Temperature	
Operation	-25 to 60°C
Storage	-40 to 85°C
Transport	-40 to 85°C
Relative humidity	
Operation	5 to 95%, non-condensing
Storage	5 to 95%, non-condensing
Transport	5 to 95%, non-condensing
Dimensions	
Width	164 mm
Height	144 mm
Depth	40 mm
Weight	150g

Table 5. Technical Data

3.6 LCP2116 - Status LEDs

Figure	LED	Color	Status	Description
	SD	Green	Off	PREOPERATIONAL mode
			On	SD Ok
			Blink 500ms	Error
			Blink 20ms	Read
	Ok	Green	On	RUN mode
			Off	PREOPERATIONAL mode
			Blinking	Error
	Pwr	Green	Off	Module power supply not connected
			On	Power supply OK
	X2X	Green	Off	Error
			On	RUN mode
			Blinking	PREOPERATIONAL mode
	PC, HMI, S1-S4	Yellow	Off	Standby
			Blinking	CPU transmitting data via the RS485
	Ethernet	Yellow	Blinking	Ethernet activity
			Green	Link has been established
			Off	No connection

Table 6. Led status

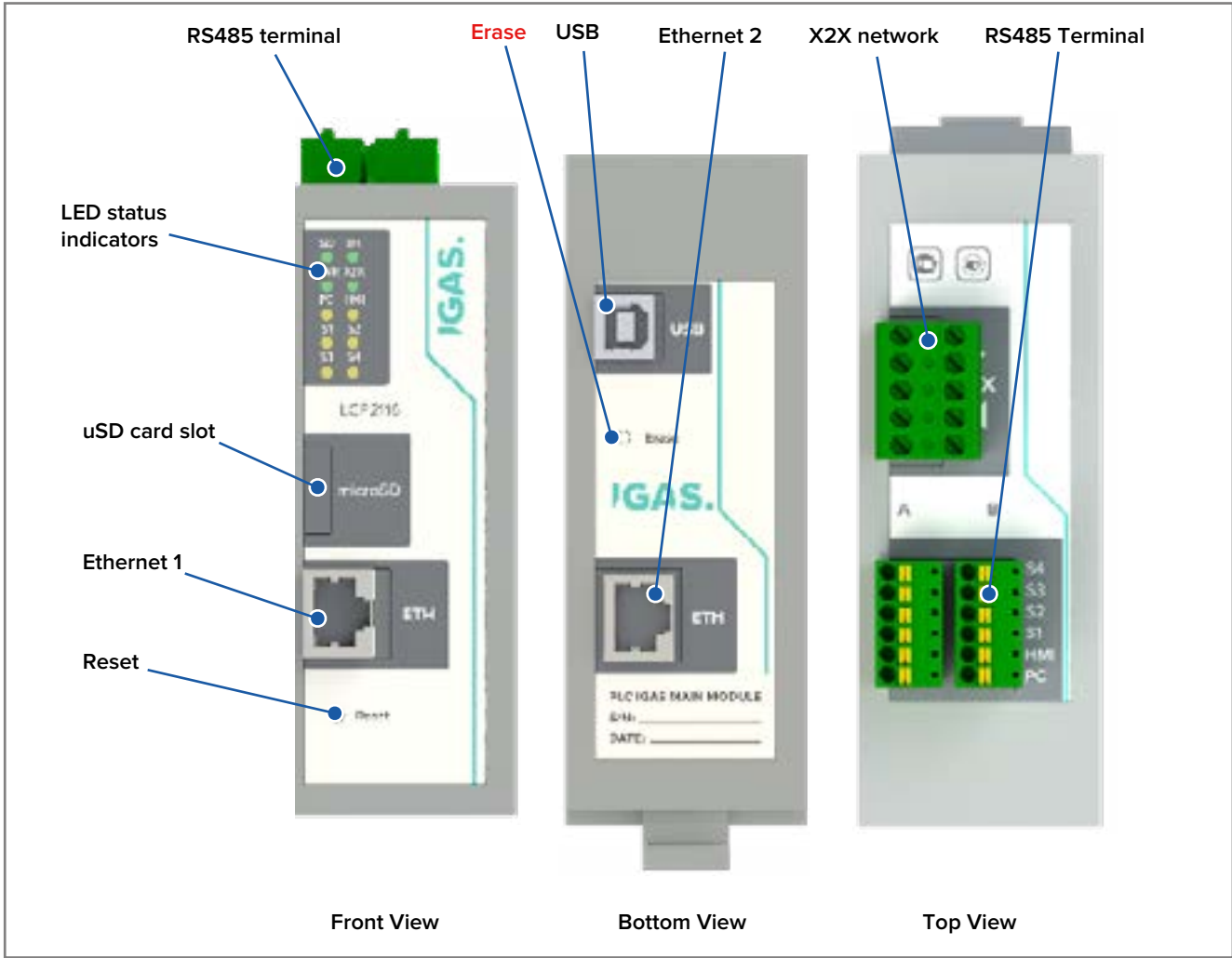


Figure 5. LCP2116 connectors

3.7 LCP2116 - Reset

The button must be pressed for less than 2 seconds to trigger a reset. This triggers a hardware reset on the CPU, which means that:

- All application programs are stopped.
- All outputs are set to zero.

The controller then boots and run.

3.8 LCP2116 - RS485 interface

Six fault-protected RS-485 interfaces with extended common-mode range. Intended to serve as an online interface for communication with the gas detectors.

Parameter	Discription
Bus-Pin fault protection	> ± 70 V
Common-Mode voltage range	(-20 V to 25 V
Electrostatic discharge	± 16 kV JEDEC
Nodes	< 256 Nodes
Signaling rate	115 kbps
Cable length	< 1500 m
Termination resistor	Yes

Table 7. RS485 parameters

3.9 LCP2116 - Project installation

Project installation is described in IGAS Automation Studio instruction for use.

3.10 LCP2116 - Ethernet interface

The interface is designed for 10BASE-T / 100BASE-TX transmission. Supports following Hardwired TCP/IP Protocols : TCP, UDP, ICMP, IPv4, ARP, IGMP.

Up to 6 simultaneous connections on each Ethernet port.

Parameter	Discription
independent sockets	8
Tx/Rx Buffers	32Kbytes
Modbus TCP	Yes
IEC 104	Yes
PRP	Yes

Table 8. Ethernet parameters

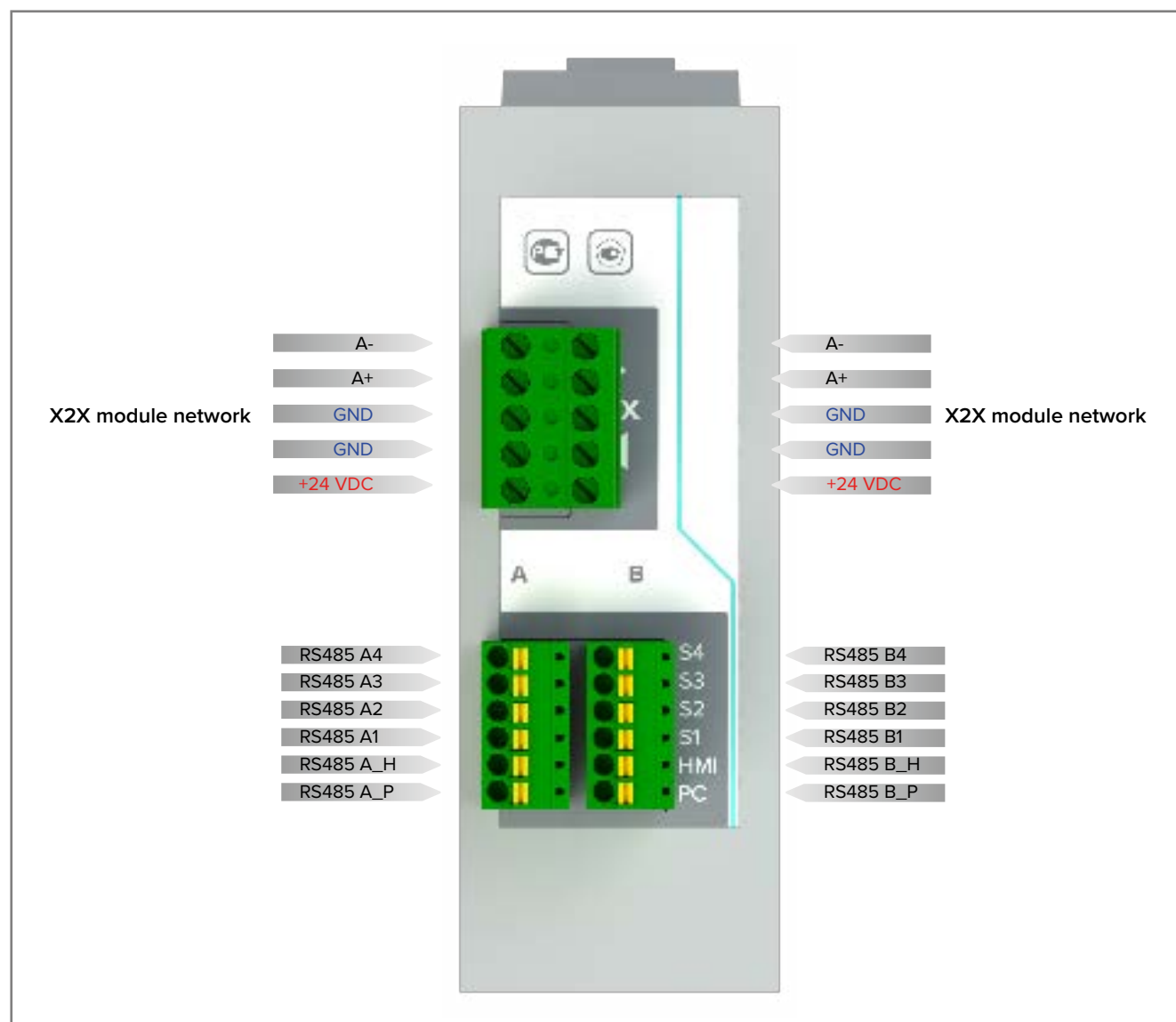


Figure 6. LCP2116 PinOut

3.11 LCP2116 - RS485 connection example

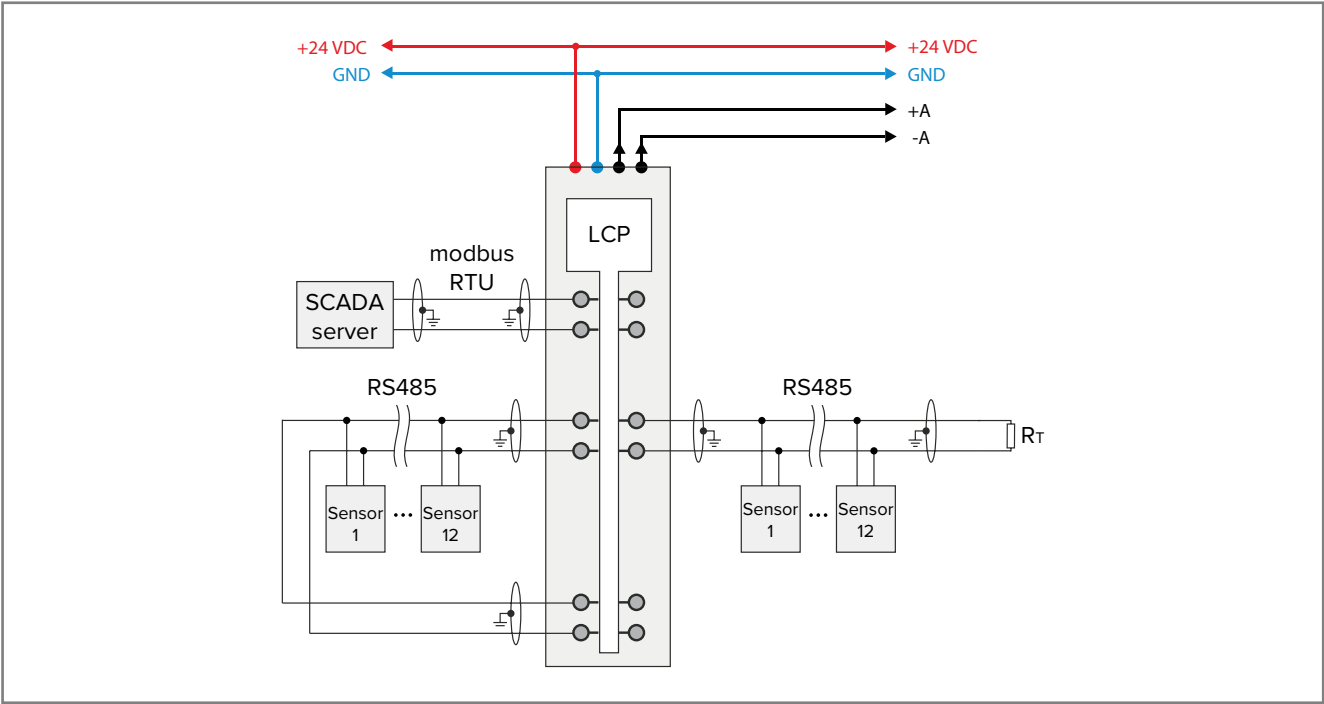


Figure 7. RS485 connection example

3.12 LCP2116 - RS485 Output circuit diagram

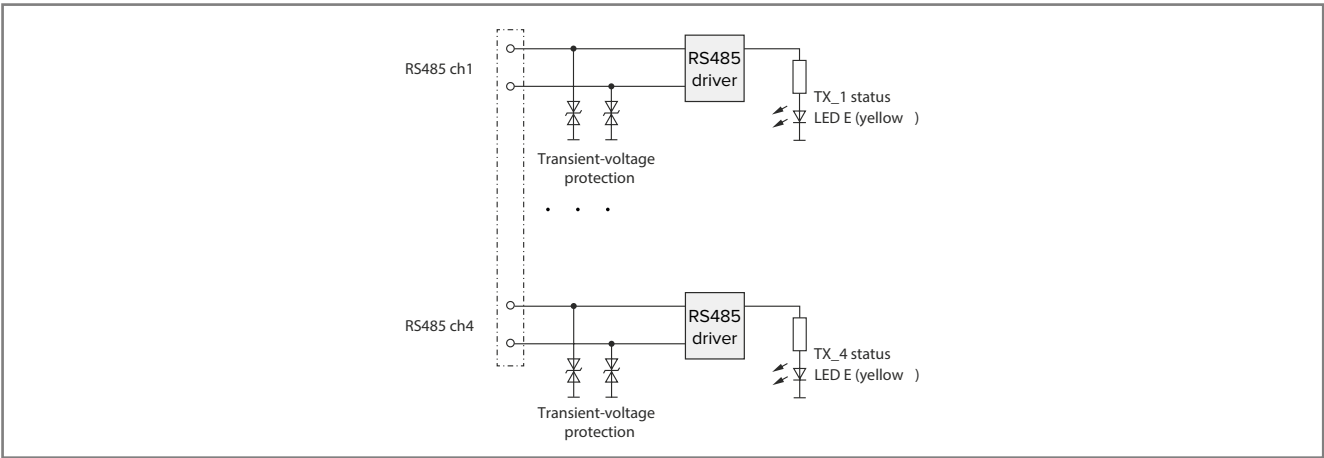


Figure 8. RS485 circuit diagram

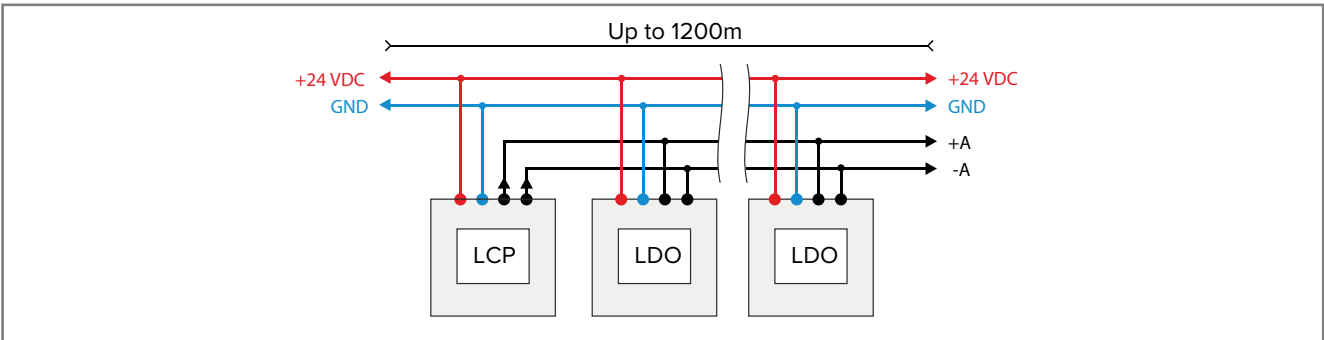


Figure 9. X2X module network

3.13 LDO1128 - Relay Output Module

General information

The module is equipped with 8 relay outputs for 3-wire connections:

- 8 digital outputs
- Relay module for 240 VAC / 30 VDC
- Switching current 2 A
- 8 normally open/closed contacts
- Push in fast terminals
- Up to 1000m remote communication



Figure 10. LDO1128 module

3.14 LDO1128 - Technical data

General information	Discription
I/O module	8 digital outputs 30 VDC/250 VAC, outputs are 3-wire, isolated
Status indicators	Operating state per channel, module status, power status
Diagnostics	
Module run/error	Yes, using status LED
Outputs	Yes, using status LED
Power consumption	Yes, using status LED
Bus	2W
Internal Outputs	-
Digital outputs	
Variant	Relay / Normally open / Closed contacts, Channels are isolated
Nominal voltage	30 VDC / 250 VAC
Max. voltage	264 VAC
Switching voltage	Max. 50 VDC / 264 VAC
Rated frequency	DC / 45 to 63 Hz
Nominal output current	2 A at 30 VDC / 2 A at 250 VAC
Total nominal current	16 A at 30 VDC / 16 A at 250 VAC
Switching voltage	Max. 50 VDC / 264 VAC
Contact resistance	Max. 100 mΩ
Switching delay	
0 → 1	≤10 ms
1 → 0	≤10 ms
Isolation voltages	
Channel - Bus	Tested at 2300 VAC
Channel - Channel	Tested at 750 VAC
Service life	
Electrical	Min. 500 000 ops. (at 2 A / 240 VAC)Channel -
Mechanical	5 000 000 ops.
Operating conditions	
Mounting orientation	
Horizontal	Yes
Vertical	Yes
Degree of protection	IP20
Ambient conditions	
Operation	-25 to 50°C
Storage	-40 to 85°C

3.15 LDO1128 - Status LEDs

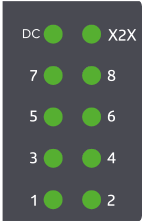
Figure	LED	Color	Status	Description
	DC	Green	Off	Module power supply not connected
			On	Power supply OK
	X2X	Green	Off	Error
			On	RUN mode
			Blinking	PREOPERATIONAL mode
	1 - 8	Green	Off	Digital output - Off
			On	Digital output - On

Table 9. Led status

3.16 LDO1128 - Relay Pinout

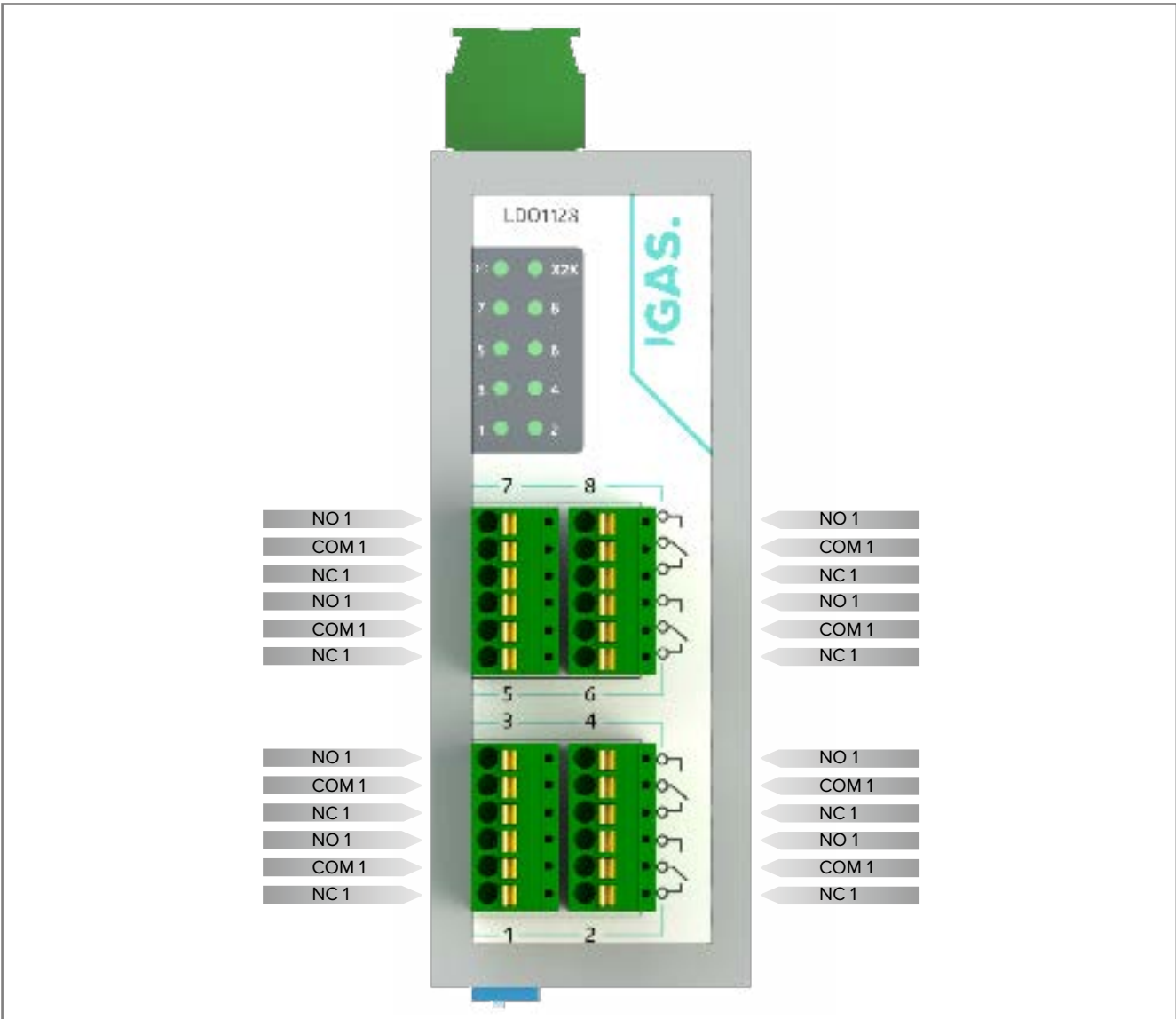


Figure 11. LDO1128 Relay pinout

3.17 LDO1128 - Supply/Network Pinout

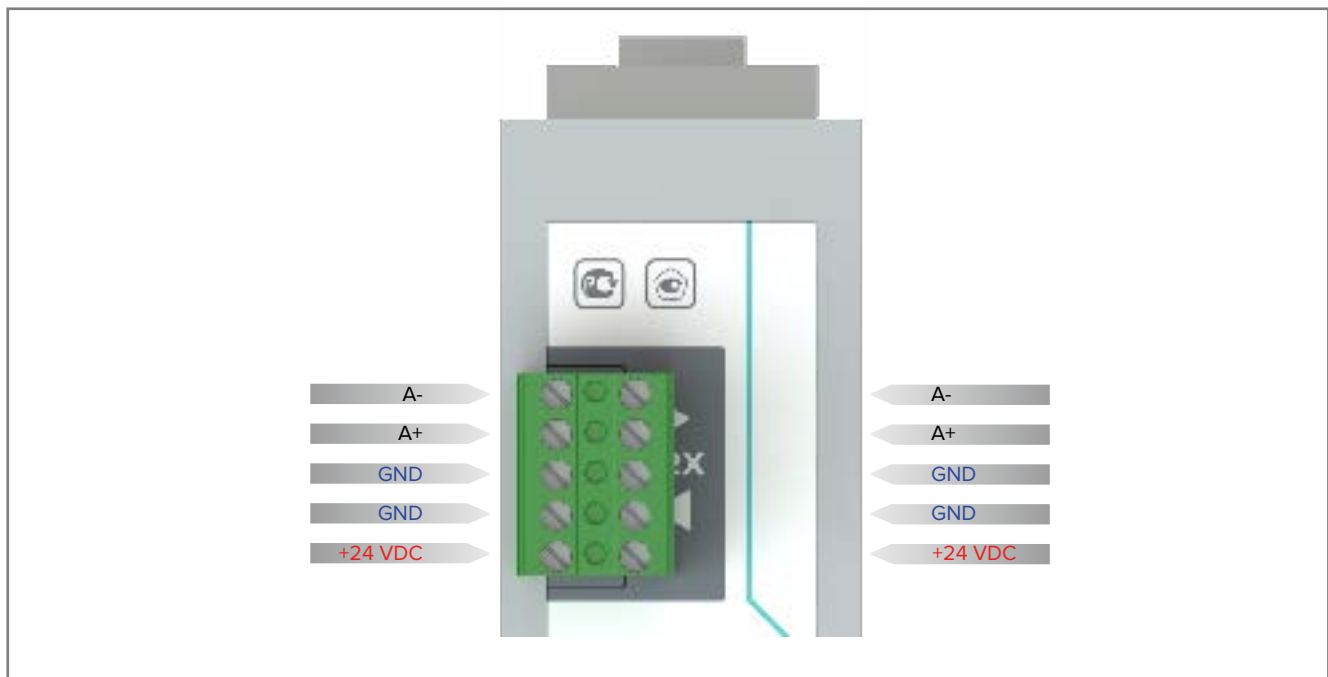


Figure 12. LDO1128 X2X PinOut

3.18 LDO1128 - Network Address Switch



Figure 13. LDO1128 Slave dip switch

3.19 LDO1128 - Connection example

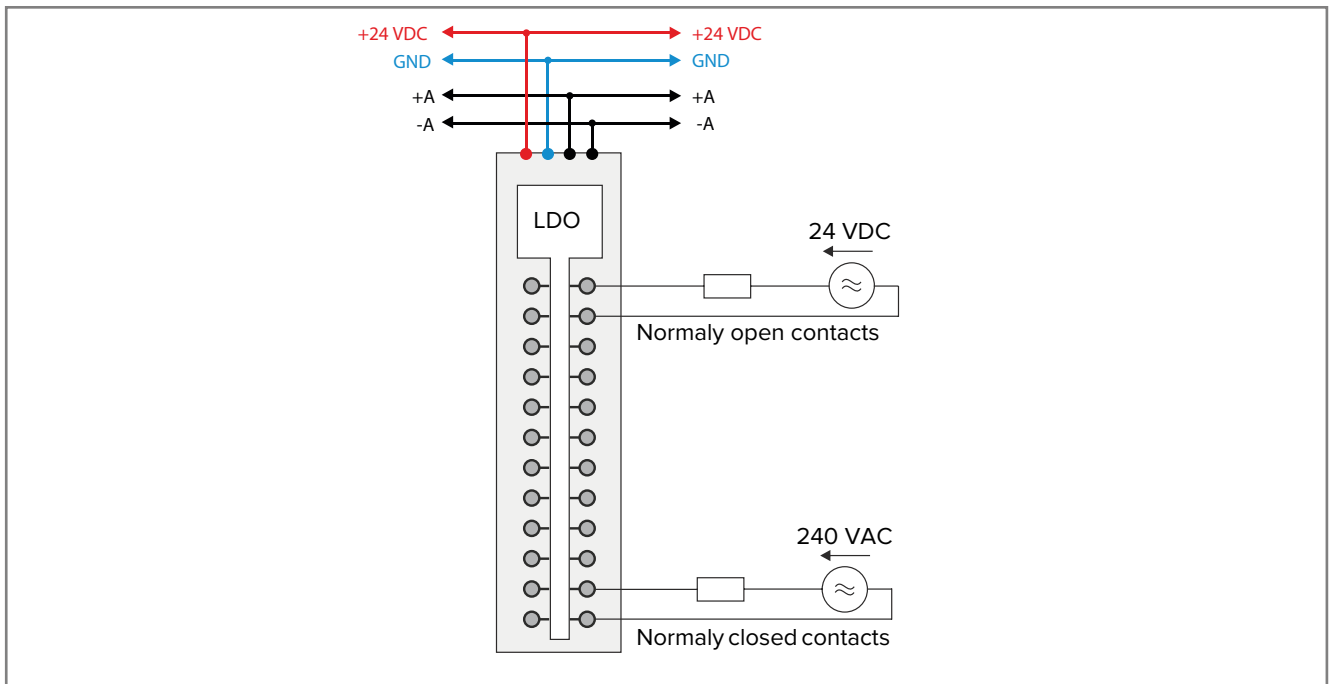


Figure 14. LDO1128 Connection example

3.20 Output circuit diagram

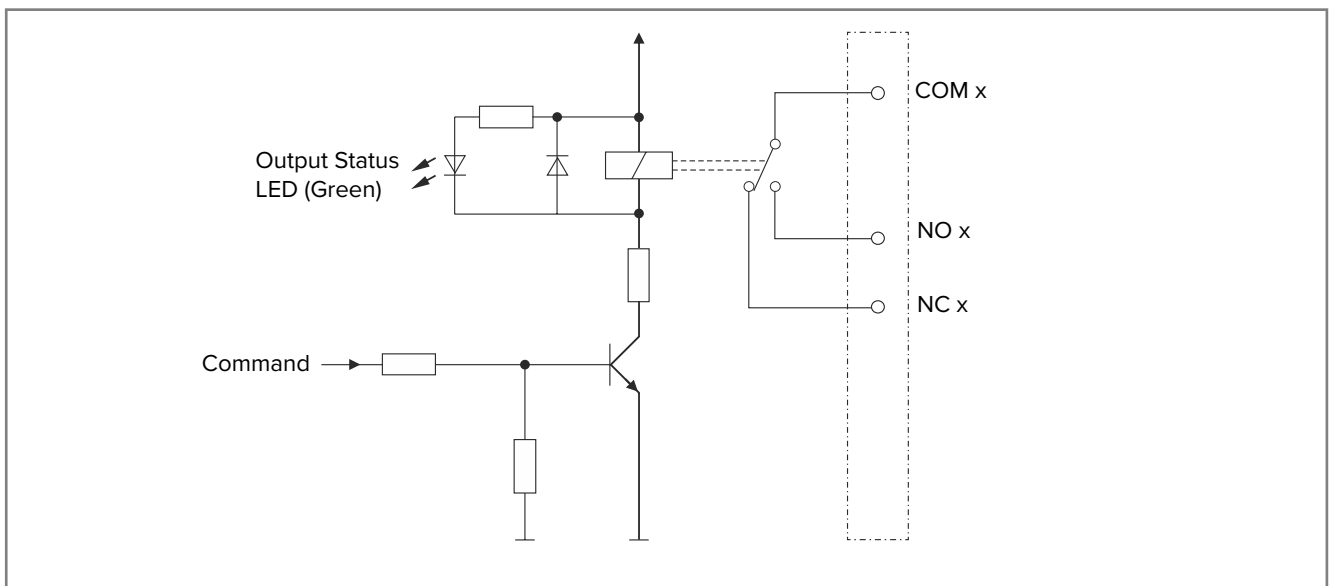


Figure 15. LDO1128 Circuit diagram

3.21 General data points

The module has general data points. These are not module-specific but contain general information such as serial number and hardware variant. Fixed modules require their data points to be in a specific order in the frame. Cyclic access occurs according to a predefined offset, not based on the register address. Acyclic access continues to be based on the register numbers.

Bit	Name	Value	Information
0	DigitalOutput01	0	Digital output 01 reset
		1	Digital output 01 set
...			
7	DigitalOutput07	0	Digital output 07 reset
		1	Digital output 07 set

Table 10. Data table

3.22 Minimum cycle time

The minimum cycle time specifies the time up to which the bus cycle can be reduced without communication errors occurring. It is important to note that very fast cycles reduce the idle time available for handling monitoring, diagnostics and acyclic commands. Recommended cyclic time > 10 ms, or use sporadic data transmission.

4 Installation and Setup Documentation

Design of a Gas Detection System for each customer carried out by IGAS. System is very flexible and configured individually. Every Gas Detection System shipped with Customer Drawing Package.

The standard customer drawing package includes:

1. System Block Diagram: Provides overview for complete system and end user utility requirements.
2. Field Unit Arrangement: Displays the external and Internal arrangement of system components and provides main system dimensions.
3. Field Unit Installation: Provides installation instructions for system and probe installation.
4. Flow Diagram: Provides complete flow diagram and logic table.
5. Interconnect Wiring: Details power and signal connections inside the customer connection enclosure.



NOTICE

System installation and wiring has to be done refer to IGAS Customer Drawings.

Strictly follow the Instructions for Use.

Use only IGAS approved parts.

4.1 Common design requirements

- Prevent unauthorised access
- Available nearest grounded chassis point
- Protect against shock, vibration, high or low temperature, overpressure, excess voltage and humidity
- Use only proper interface connections to external devices
- Separate power and signal cables
- IGAS recommend to use shielded cables for detector wiring
- Use Cat-5e for Ethernet (upto 100 meters)
- Prevent output overloads
- RS485 cable impedance - 120 Ohm

4.2 Wiring

All Input and Output terminal blocks with Push-in connection technology. Insert rigid conductors or conductors with ferrules from 0.25 mm² into the conductor shaft. The contact spring opens automatically and provides the required pressure force against the current bar. When installing smaller conductors from 0.14 mm², use a standard screwdriver to actuate the contact spring.

Double check wire insulation. Replace if damaged.

Do not stress cables.

Any damaged terminal has to be replaced immediately.

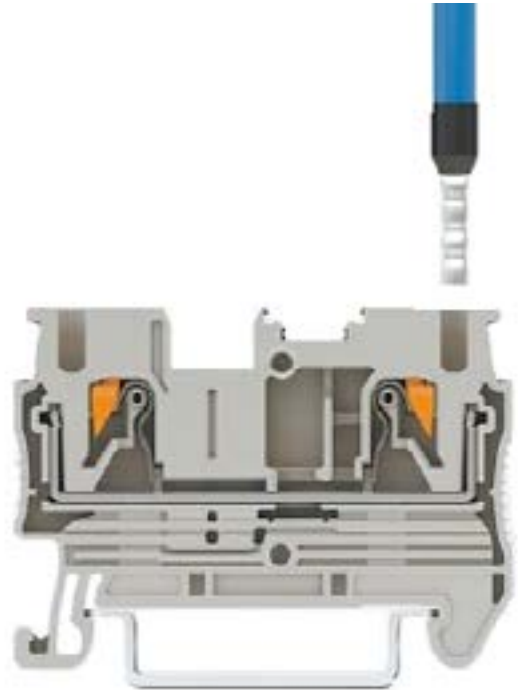


Figure 16. Push In technology

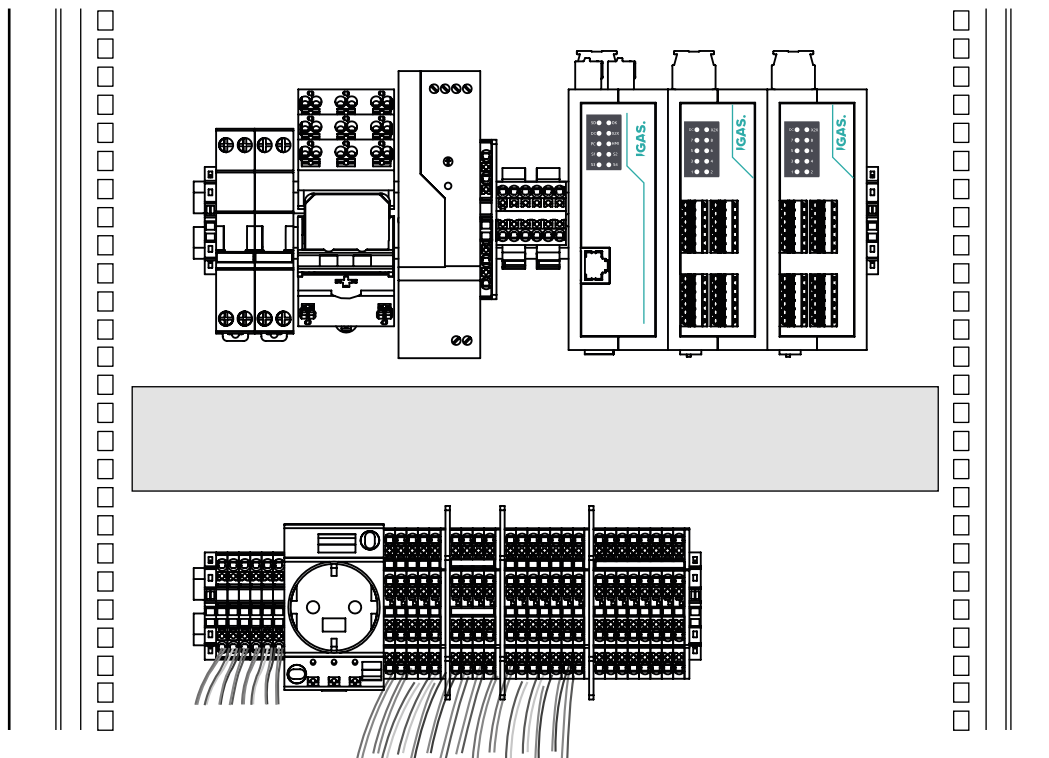


Figure 17. Wiring

5 GDS User Guide

5.1 User Interface General



CAUTION

Opening the enclosure may expose live electrical circuits. Touching exposed terminals or wires may cause death or serious injury. Do not operate system in normal mode with the door insecure.

Take alternative site safety precautions while power is off.

The User Interface panel has:

A colour Touch screen for normal system operation, maintenance and configuration

- Power, Alarm / Fault and Inhibit state indicators
- Active Channel (01 to 40) status indicators
- IP addresses of connected servers via Ethernet

Further System Interfaces consist of:

- Silence sounder button
- 16 configurable relays in the basic configuration
- Detector alarm, System Alarm, Gas level warning and alarm outputs for visual and audio alarms
- An SD Card slot for data logging and firmware/software updates
- X2X network for expansion modules
- Remote Interface networking via RS485 port
- Remote Modbus TCP/IP or IEC104 protocol

5.2 Touchscreen

The Touchscreen is touch only; it has no swipe or pinch gestures.

The Touchscreen has two access levels: View mode is available to all users while Configuration, Maintenance and System Test functions are password protected.

5.3 Switching On and Off

Before switching power on, ensure that the system has been commissioned or that a qualified person has checked that the wiring is safe and conforms to local regulations. Also check that the battery isolator switches are 'On' and that an SD Card is fitted. Check that the optional expansion unit battery switches are on too.

Switch on power at the circuit breakers and wait for the system to initialise. (The system start up sequence may last for up to 15 sec depending on the number of channels in use.)

GDS will first display the main screen, indicating the current status of the detectors. The screen shows installed sensors but it may be blank depending on your system status

ICON	Discription
	Settings Menu Options
	Event History
	Event History
	Sounder enabled
	Sounder silenced
	No one Logged in
	Admin Logged in
	Battery Ok
	Battery is running low (blinking)
	Latched Alarm Events (blinking)

Table 1.1 Home screen menu buttons.

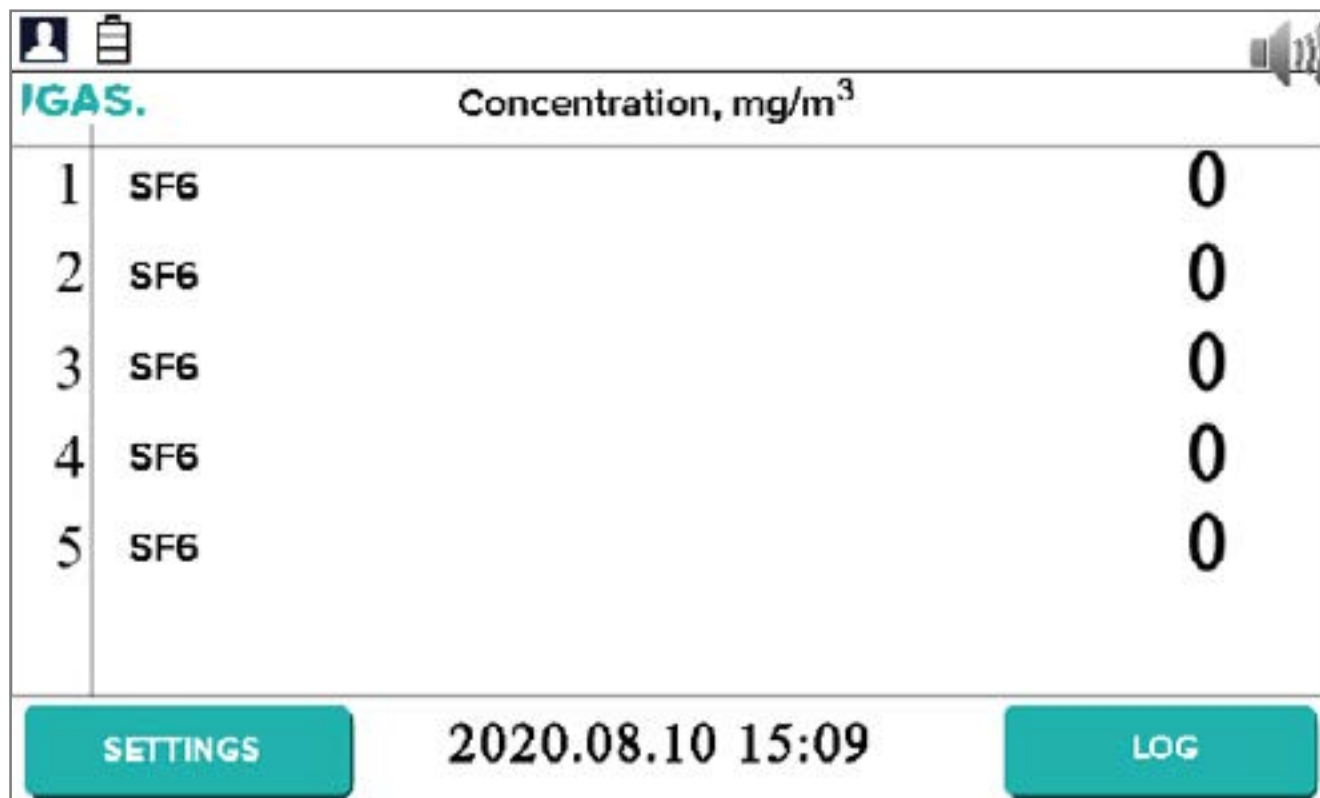


Figure 18. Home screen.

5.4 Menu Items and Access Levels

The table below details the menu items and access levels for the User Interface. The Administrator can edit everything. Operator can view, and acknowledge events. A user can log in by touching the log in icon, and inputting a valid password.



NOTICE

Authentication reset follows 15 minutes of inactivity. For security reasons do not leave the Touchscreen

● = Allowed, ○ = Denied, ↗ = sub-menu, ↘ = sub-sub-menu

Menu Item	Operator	Admin
Settings	●	●
↗ Ports	○	●
↗ Relay	○	●
↘ Gas Relays	○	●
↘ Fault Relays	○	●
↗ Date And Time	●	●
↗ Service	○	●
↘ Modules Status	●	●
↘ Emulation	○	●
↘ Toolbox	○	●
↘ Relay Test	○	●
↘ Redundancy	●	●
↘ IEC 104	○	●
↗ Delete LOG	○	●
↗ Detector Settings	○	●
LOG	●	●

Table 12. User / Component Matrix

5.5 SD Card Usage

The SD card is used to store the system event history and configuration. GDS logs all events. System accepts standard size SD cards of 2 to 32 GB (FAT32). SD cards must be formatted on first insertion.

5.6 Normal Operation

During normal operation:

- System will collect sensor data every 100 ms from ports and modules.
- The Cause and Effect matrix will be evaluated immediately on event, and commands sent accordingly to the appropriate output channels.
- Any change in status of channel will be reported to the User Interface and logged in the event history.
- Events (Alarms, Faults, Inhibits, etc.) will be reported to the User Interface and logged in the event history.
- Any fault within the system will activate the System Fault relay.
- Any failure of the safety function i.e. due to lost sensor connection will activate the Detector Fault relay.

5.7 Touchscreen interactions

The colour Touchscreen is activated using a finger. Do not use sharp or abrasive objects as they may cause irreparable damage.

All interactions are single tap (no gestures or swipes). Some actions open a new Window; depending on the window type, they can be closed by touching the X or [Cancel] button.

5.7 User Interface Screen

The icons in the navigation toolbar are used to toggle through the User Interface options, as shown in the tables below.

Ok	Warning Event	Alarm Event
----	---------------	-------------

ICON	Discription
BACK	Go upper menu screen

	Concentration, mg/m ³
1 SF6	24
2 SF6	16
3 SF6	2
4 SF6	250
5 Disabled	

	Concentration, mg/m ³
1 SF6	0
2 SF6	3200
3 SF6	0
4 SF6	732
5 SF6	0

	Concentration, mg/m ³
1 SF6	5120
2 SF6	3200
3 SF6 Lost connection	
4 SF6	0
5 SF6 Error	

ID	Date	Time	Status	Description
004	08.08.20	19:47	F	Detector 0, Fault
005	08.08.20	19:47	F	Detector 0, Fault
002	08.08.20	19:47	F	Detector 0, Fault
008	08.08.20	19:47	F	Detector 0, Fault
007	08.08.20	19:47	F	Detector 0, Level 1
005	08.08.20	19:47	F	Detector 0, Level 1
000	08.08.20	19:47	F	Detector 0, Level 1
000	08.08.20	19:47	F	Detector 0, Level 1

Touchscreen LOG is a kind of a book with lots of pages. Delete - move log to new next page. Recove - return back to previous pages.

5.7 Navigation – Menu

User must have the appropriate access level password to enter Configuration and System Test options.

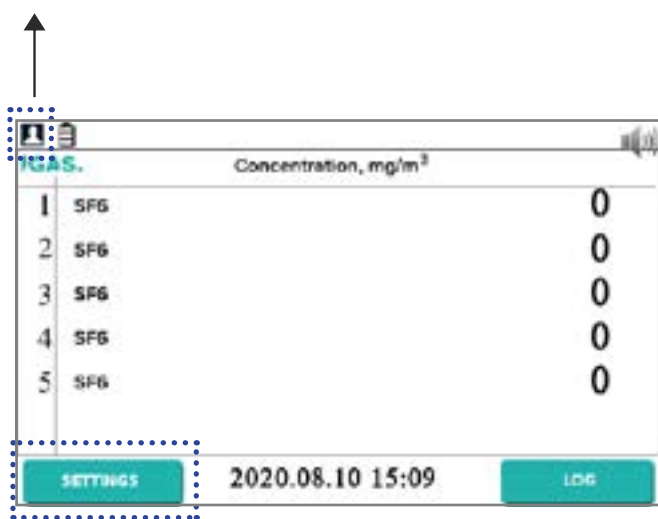


Figure 20. Menu access

5.8 View Active Alarms

Active alarms can be viewed at Home screen. Touch the log button icon to display a list of the events, starting with the most recent event. To acknowledge active an alarm, press the **silence the sounder** button which will silence the alarm for 10 minutes.



Figure 21. Events color



NOTICE

Note: Check for new alarms if the sounder restarts after you have acknowledged it.unattended while logged in.

5.9 Event Information

Detailed information about active events can be viewed in LOG screen by touching **LOG** button on the bottom toolbar. The screen will then show a list of events, date, start and finish time of events. To clear log - push **delete** button

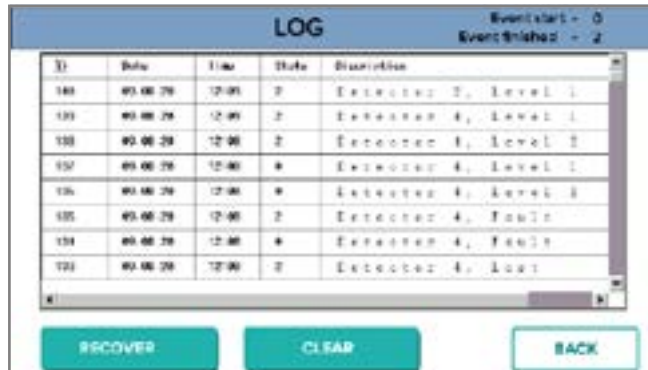


Figure 22. Log screen

5.10 Acknowledging Events

Active and latched events can be acknowledged by pressing the LOG button. This will acknowledge ALL active events and the flashing will stop.

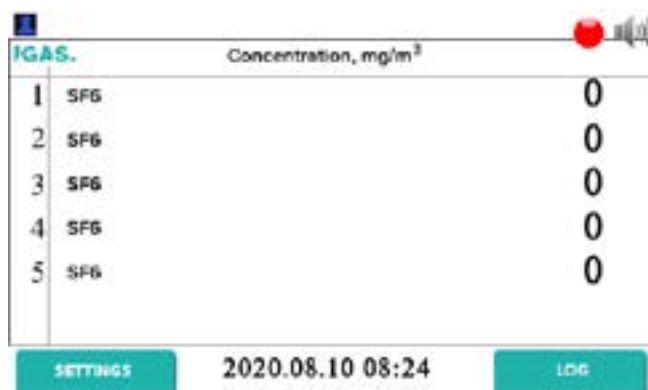


Figure 23. Latched events

6 System configuration

6.1 General Information

For IGAS GDS use the IGAS Automation Studio Software. The software is a Microsoft Windows based program that requires USB B cable and a compatible PC. The software and interface unit are available from IGAS. See the Instructions for Use supplied with the software for full instructions.

Several settings can be changed using the built-in configuration system.

6.2 RS485 ports

4xRS485 Serial ports available. PORTS screen used to bind RS485 port to proper output terminal. PORTS can be viewed in [Settings>PORTS](#).



Figure 23. Serial port screen

6.3 Output Relays

From the HOME screen **Settings>Relays**.

From the Relays screen:

1. Select digital output module and detector address.
2. Set 1st (Yellow) and 2nd (Red) gas levels
3. Set Alarm relays
4. Save configuration.

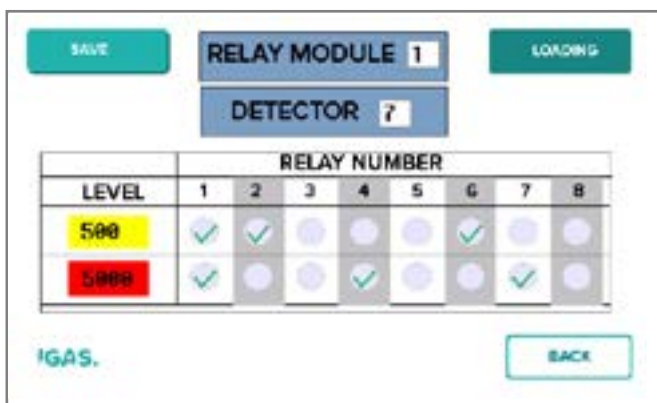


Figure 24. Relay settings

6.4 Date and Time

From the HOME screen **Settings>Date And Time**.

1. Set data.
4. Save configuration.

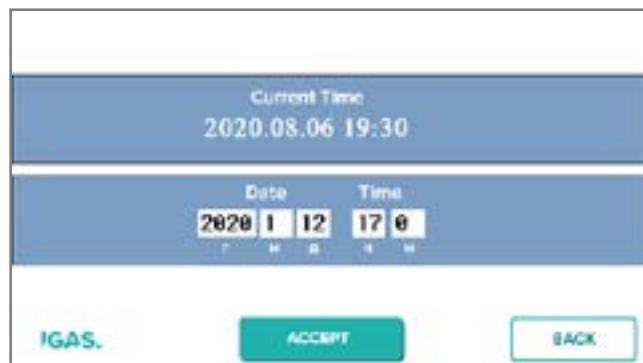


Figure 25. Date and time screen

6.5 Inhibiting Detectors

To Inhibit Detector:

- 1) Log in as Administrator
- 2) Go [Settings>Detector Settings](#)
- 3) Choose detector number and *Inhibit or Enable it*
- 4) Save settings

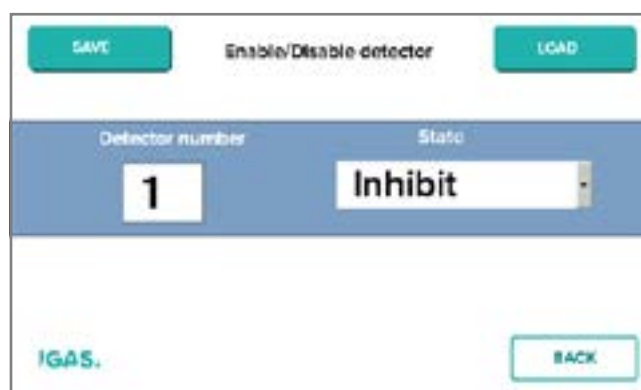


Figure 26. Enable / Inhibit screen

6.6 Ethernet Protocol

The Ethernet Interface supports two protocols and 6 clients. IP, subnet, gate etc, can be configured in IGAS Automation Studio. Touchscreen allow to choose protocol IEC104 or Modbus TCP.

From the HOME screen **Settings>Service>IEC104**.

1. Choose protocol.
2. Accept choice
3. PLC will reboot. Wait for loading.



Figure 27. Ethernet screen



NOTICE

If system connected to SCADA via IEC 104, no need to manually set date and time. SCADA sync it automatically. Otherwise it has to be done manually



NOTICE

All valid connections listed in Connected Servers table.

7 Maintenance and Scheduled Testing

This chapter describes how to perform common maintenance operations and system testing.



CAUTION

Test outputs will be generated and relays will be activated during Testing.



NOTICE

While an Operator could Force a channel, a Service password is required to carry out tests and restore the system to full operational status.

7.1 Routine Testing

It is normal practice to sound the site alarms periodically, both to check they work and to accustom personnel to the sound they make.

As it involves inhibiting emergency systems during testing, it might be more prudent to test the GDS relays only on a planned monthly basis.

Visually inspect all cables for loose connections, security of attachment, and signs of damage.

Wipe the Touchpoint Plus Controller with a soft damp cloth or use PC Screen cleaner.

Check for dust contamination.

Periodically exercise the outputs (see the following chapters) to confirm their functionality.

7.2 Exercising the Relays

Relay output channels can be tested by forcing the relays to an active state:

1. Ensure that relay activated output systems (i.e. emergency deluge / sirens etc.) are isolated before starting the test.
2. Login as an Administrator or Service.
3. Touch **Settings>Service>Test Relay**.
4. Touch the active (yellow) Relay Output Channels to be tested, and toggle them on.
5. Check the selected relay output contact power switching using a suitable test tool or voltmeter.
6. Touch [Finish]>[Cancel] and all selections will turn off.
7. Proceed to the next step.



CAUTION

If the relays are switching mains voltages, hazardous live terminals may be present.

Ensure that all mains current is isolated and dissipated before touching live terminals.

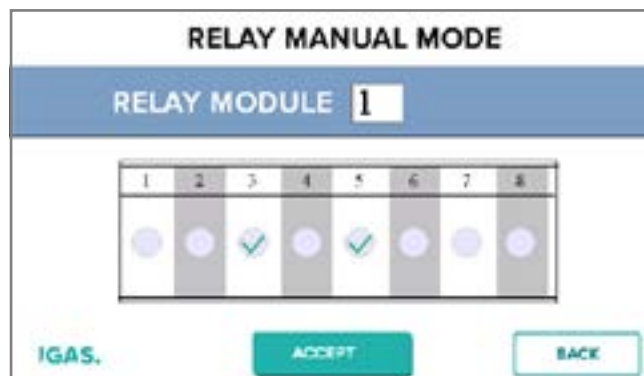


Figure 28. Force relay screen

7.3 Check system reaction.

GDS reaction on every event (gas level alarms, faults etc) can be tested by emulation of the event.

1. Touch **Settings>Service>Emulation**
2. Choose detector number
3. Choose event
4. Activate and accept.

Ensure that relay activated output systems (i.e. emergency deluge / sirens etc.) are isolated before starting the test.

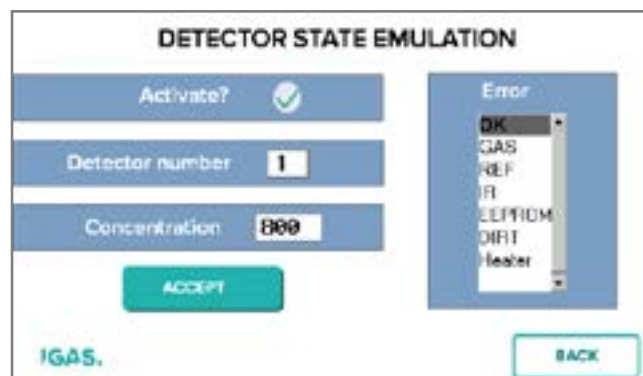


Figure 29. Event emulation screen



CAUTION

Emulation will generate digital output signals and signals in corresponding registers via Ethernet and RS485 to SCADA.

7.4 Periodic Scheduled Testing.

Periodic Scheduled Testing should only be carried out by an IGAS authorised technician or a qualified trained person. Apart from the safety considerations, IGAS's field service technicians will be able to minimise downtime should a fault become apparent during testing.

8. Servicing

Repairs, Replacements and Upgrades should only be carried out by authorised IGAS technicians or by qualified persons trained in accordance with the IGAS requirements. Failure to follow this advice will invalidate your warranty and may lead to death, injury, or irreparable damage.

8.1 Back up Battery Maintenance

Replace the battery only with battery type CR2032.
Use of other batteries may present a risk of fire or explosion.
Batteries shall only be fitted, removed or replaced in non-hazardous (safe) areas.



CAUTION

Lithium batteries may cause severe injury or death if put in the mouth or swallowed.
Always keep them away from children and dispose of them as hazardous waste.



CAUTION

According to local laws and regulations the battery should be disposed of separately from household waste. When the battery reaches its end of life, take it to a collection point designated by local authorities.

To replace the back up battery:

1. Ensure that the environment is safe and the atmosphere is below flammable levels.
2. Switch GDS off and isolate its power supplies.
3. Disconnect plugs from PLC unit.
4. Unmount PLC from din rail
5. Unscrew the four locking screws and open the enclosure.
6. Replace the battery with cr2032.
7. Close the enclosure and secure with the four screws.
8. Mount PLC on din rail and connect all necessary plugs
9. Switch power on and restore GDS to normal operation.
- 11) Look at the power icon to check battery state.

9 Troubleshooting



CAUTION

Read Chapter 2 Safety and other related warnings and cautions before carrying out any work on the gas detection systems.

Problems are signalled by an error shown on the control panel display. The error is recorded in the event log and the screen can be cancelled once the fault is corrected.
The error messages are explained in Fault table.
Contact IGAS if an error appears repeatedly, if it cannot be cleared, or if it is not listed in fault table.

9.1 Calling for Technical Support.

Please contact IGAS Technical Support if your problem cannot be resolved or if you need additional help. You can find our contact details on the rear cover of this manual.

To assist the IGAS Helpdesk in giving you technical support, it would be helpful to have the following information ready:

- Your Company contact details and a contact name and department.
- The system Serial Number. These can be found on the system identification label.
- Error messages and code numbers that were shown on the display screen.
- Details of the difficulty that you are experiencing.
- Status of the system (Inoperative, partially operative, still functioning etc.)

To assist with the fault diagnosis you may be asked to send a copy of your event log by email.

9.2 Fault Codes

Display in Event History	Description
Detector lost	Detector does not respond. Please check the connection.
Memory fault	uSD memory card fault
uSD missing	uSD memory card is not inserted
Detector fault	Remote fault notified by a detector. Check detector manual for more info
Relay Output Error	Relay not controlled
Battery Fault	Backup battery fault or running low
Configuration Fault	Configuration file corrupted
Loop broken	RS485 redundant loop is broken

Table 13. Fault codes table

9.3 Modules Status

Information about PLC and modules can be viewed in [Settings>Service>Modules Status](#). A list of all configured modules will be displayed.

Green LED - module OK

Red LED - Error

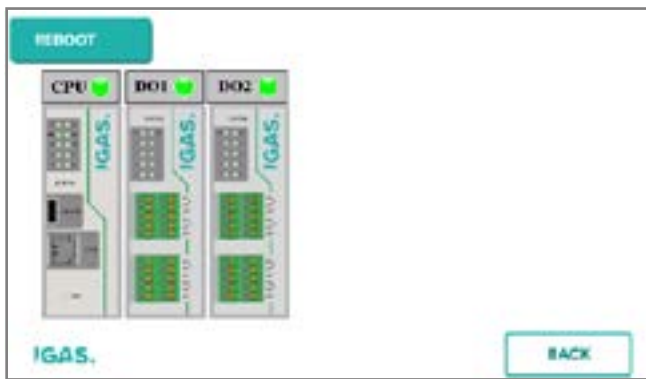


Figure 30. Modules status screen

9.4 Redundant loops

Information about redundant RS485 loops [Settings>Service>Redundancy](#).

Loop state status:

Green LED - loop OK

Red LED - Loop broken

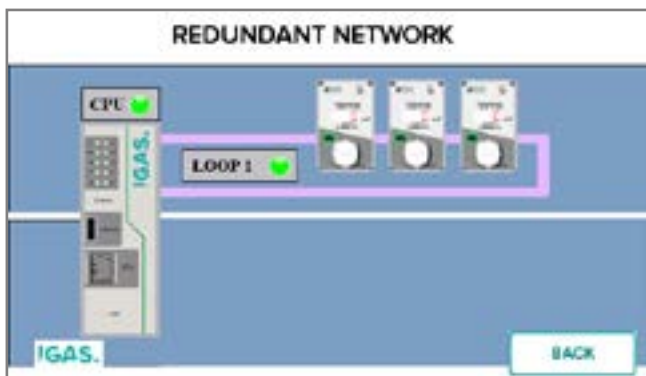


Figure 31. Redundancy screen

9.5 Modules Replacement

IGAS GDS is a modular system.

This modularity results in a system that combines the advantages of hot pluggable electronics and extra bus slots for added options.



CAUTION

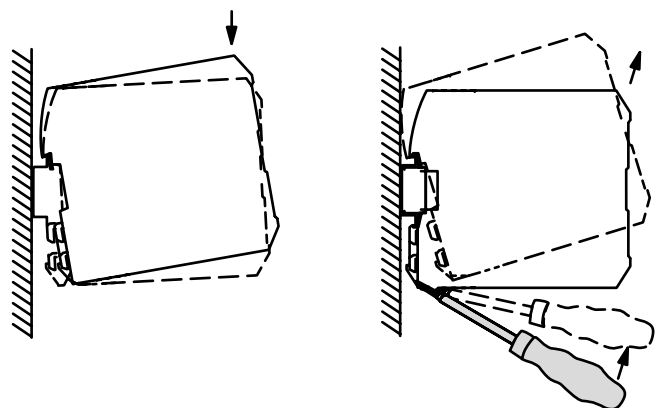
Prevent short circuit while hot swaping module.

Removing

1. Unplug connectors from module
2. Use a suitable screwdriver to release the locking mechanism on the snap-on foot of the device. Hold onto the device by the housing cover and carefully tilt it upwards.
3. Carefully lift the device off the DIN rail connector and the DIN rail.

Mounting

1. Set correct module bus address (Address should be the same as it was on removed module)
2. Connect the DIN rail connectors together for a connection station.
3. Push the connected DIN rail connectors into the DIN rail.
4. Place the device onto the DIN rail from above. Ensure the device and DIN rail connector are aligned correctly.
5. Holding the device by the housing cover, carefully push the device towards the mounting surface so that the device is securely fixed onto the DIN rail connector.
6. Once the snap-on foot has been audibly snapped onto the DIN rail, check that it is fixed securely. The device is only mechanically secured via the DIN rail.
7. Re-connect unplugged connectors.
8. During startup, check that the device is operating, wired, and marked correctly.



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